Replication of "'In- Group Love' and 'Out- Group Hate' as Motives for Individual Participation in Intergroup Conflict: A New Game Paradigm" by N Halevy, G Bornstein, L Sagiv (2008, *Psychological Science*)

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In their original paper, "'Ingroup Love' and 'Outgroup Hate' as Motives for Individual Participation in Intergroup Conflict: A New Game Paradigm," Halevy, Bornstein and Sagiv (2008) discuss why participants follow certain behaviour patterns that negatively impact on themselves during an intergroup conflict. We chose to replicate this particular study because it is close to the first author's research interests in social psychological applications of economic game paradigms (e.g. Thomae, Zeitlyn, Griffiths & Van Vugt, 2013) but also because it fits well into both authors' broader area of expertise in group processes and intergroup behaviour.

Halevy et al. implemented both the intergroup prisoner's dilemma (IPD) game and the intergroup prisoner's dilemma-maximizing difference (IPD-MD) game to distinguish whether participants are seeking to harm the outgroup or help the ingroup in engaging in such "self-sacrificial behaviour." Harm during intergroup conflict in this study is operationalised as allocating tokens to a pool which will not only financially benefit one's own group (ingroup) but simultaneously financially penalise the other experimental group (outgroup).

The IPD game asks participants to either help their ingroup while simultaneously harming the outgroup or to take no action at all for personal gain. Participants in this game can contribute experimental tokens to Pool B, which adds £0.60 per token to the ingroup pool (£0.20 for each player), while simultaneously reducing the outgroup pool by £0.60 (£0.20 from each player) per token. The IPD-MD game, on the other hand, allows participants to make choices that either help their ingroup without harming the outgroup, help their ingroup while simultaneously harming the outgroup, or to take no action at all for personal gain. In the IPD-MD game, participants can either allocate tokens to Pool B, with the same consequences as described above, or can alternatively allocate tokens to Pool A, which adds £0.60 per token to the ingroup pool (£0.20 for each player), without reducing the payoff for the outgroup. Both conditions also allow for 'selfish' behaviour, such that participants can keep any number of tokens, which in this case will neither contribute to the ingroup pool nor deduct from the outgroup pool but will pay £0.40 per token for the individual player.

In this replication of Halevy et al. (2008), we sought to exactly reproduce the methodology used in the original paper. The authors tested whether the IPD-MD game would reveal a specific, unique motivation for behaviour compared to the IPD game. Additionally, they tested whether intragroup discussion would affect the significance of the motives. We focus on their finding that

"the contribution rate to pool B [between group] in the IPD-MD game was significantly lower than the contribution rate to pool B (the only pool) in the IPD game, F(1, 76) = 109.74, p = .000." (Halevy et al., p. 408)

Method

Power Analysis

The replication team reviewed the article and selected the following effect as the key finding for replication:

> "The most important finding concerns how contributions were divided between pools W [within-group] and B [between-group] in the IPD-MD game. As Figure 1 shows, contributions were made almost exclusively to the cooperative, within group pool. In the no-communication condition, participants contributed an average of 47% of their endowment to pool W and less than 6% to pool B.3 Following within group communication, contributions to pool W increased to 68%, t(29) = -2.756, p = .010, whereas contributions to pool B remained low, at 4%." (Halevy et al., p. 408)

However, during initial review of the replication study, the original first author suggested that the effect reported in footnote 3 was the key result from that study:

> "The contribution rate to pool B in the IPD-MD game was significantly lower than the contribution rate to pool B (the only pool) in the IPD game, F(1, 76) = 109.74, p = .000." (Halevy et al., p. 408)

The original study author provided the following rationale:

> "The reason that the high contribution rates to the within group pool in the IPD-MD game are identified as 'the most important finding' is because they reduce intergroup conflict. A direct measure of the degree to which intergroup conflict was reduced appears in footnote 3 (i.e., the significantly lower contribution rates to the harmful between-group pool in the IPD-MD game as compared with the IPD game)." (Correspondence with Dr. Halevy)

Thus, at the author's suggestion, we selected the latter effect as the primary test, and the effect identified by the replication team was identified as a secondary test.

To calculate the sample size necessary for an adequately powered replication, an effect size of partial $\eta^2 = .59$ was determined using Daniel Lakens's effect size calculation table. This was then entered into G*Power, with the test family "F tests" and the statistical test "ANOVA: fixed effects, main effects and interactions" selected. The numerator df was entered as 1 and the number of groups as 2; the effect of interest was the main effect of game type on B pool

contributions. This produced a result of 15 groups being necessary for a replication with 99% power, or 45 participants.

Planned Sample

Despite the power analysis yielding 15 groups as sufficient for 99% power, the present authors planned to collect 40 groups of participants. Groups were three people each for a total of 120 participants.

This sample size was chosen because the suggested 15 was significantly lower than the original paper's 80 groups. Increasing the sample size to 40 groups would lead to more stable means and variances while still being conscious of the limitations of the small participant pool at University of Winchester.

Prior to data collection, it was decided that should 40 groups be collected with more participants remaining in the participant pool and funds still allowing for continued collection, more groups would be included until resources ran out.

Each group of three would be made up of participants of the same sex (ie. 3 females or 3 males).

Participants were recruited through a university participant pool, which also advertised remuneration. All participants were paid, depending on the outcome of the game, a sum between £2 and £14.

Materials

Below are rules for the IPD and IPD-MD games. Participant instructions can be found in the Files section of the Materials component on the OSF. The use of materials below was followed precisely:

"The game is played by two groups, with 3 members in each group. Each player receives an endowment of 10 tokens and can contribute any number of these tokens to the group's pool. For each token contributed by a member of the ingroup, each of its members, including the contributor, gains 1 money unit (MU) and each member of the outgroup loses 1 MU. For each token kept, the player is paid 2 MU. Because the individual's return from contributing a token is 1 MU, but his or her cost is 2 MU, the dominant individual strategy —the strategy that yields the highest personal payoffs regardless of what all the other ingroup and outgroup members do—is to contribute nothing (i.e., defect). However, because contributing a token generates a total of 3 MU for the group while costing only 2 MU, the dominant group strategy—the strategy that yields the highest payoffs for each group regardless of what the other group does—is for all group members to contribute all their tokens."

"To disentangle any motivational ambiguities, we introduce a new paradigm, called the intergroup prisoner's dilemma–maximizing difference (IPD-MD) game. Like the IPD game, the IPD-MD game involves a competition between two 3-member groups. Each group member receives an endowment of 10 tokens, each worth 2 MU, and can decide how many of these tokens to contribute. Unlike in the IPD game, however, contributions can be made to two different pools. Contributing a token to the within group pool (pool W) increases the payoff for each ingroup member, including the contributor, by 1 MU, without affecting the outgroup. Contributing a token to the between-group pool (pool B) increases the payoff for each ingroup member, including the contributor, by 1 MU, and at the same time decreases the payoff for each outgroup member by 1 MU. The choice between pools W and B is what reveals the specific motivation underlying each individual's behaviour." Figure 1 depicts a visual display of the 2 x 2 experimental design.

		Condition: Communication	
		No communication	Communication
Condition: Type of Game	IPD	**Condition 1:** *IPD* *No communication*	**Condition 2:** *IPD* *Within group communication*
	IPD-MD	**Condition 3:** *IPD-MD* *No communication*	**Condition 4:** *IPD-MD* *Within group communication*

Figure 1: Experimental design (2 (Type of Game) by 2 (Communication).

Procedure

The procedure below was followed precisely.

"Participants arrived at the laboratory in cohorts of 6 and were randomly assigned to one of the four conditions (IPD or IPD-MD with or without intragroup communication). The participants in each cohort were randomly divided into two 3-person groups."

The 6 participants were all of the same sex; there were either two male groups or two female groups.

Each group was escorted into separate private rooms, where [group members were] "given instructions concerning the rules and payoffs of the relevant game. The instructions were phrased in neutral language (e.g., the pools were labelled ''A'' and ''B,'' with no reference to ''cooperation'' or ''competition'') and explained how each player's own decision and the decisions of the other players would affect the payoffs."

"Participants subsequently answered a short quiz that tested their understanding of the game's rules. Each participant received an endowment of 10 tokens and had to indicate on a ''decision form'' how many tokens he was keeping, how many he was contributing to pool W (the within group pool in the IPD-MD game), and how many he was contributing to pool B (the between-group pool in the IPD-MD game and the only pool in the IPD game). Each token that was contributed to either pool W or pool B paid [£0.20] to each ingroup member, including the contributor. Each token that was contributed to pool B also subtracted [£0.20] from the payoff to each outgroup member. For each token a player kept, he was paid [£0.40]. In addition, each player was paid a flat bonus of [£6] to ensure that payoffs would be positive. Thus, participants could earn between [£2] and [£14] depending on the outcome of the game."

"Participants in the no-communication condition made their decisions without communicating with the other ingroup members. Participants in the communication condition met with the other members of their team for 5min of free discussion and then returned to their rooms to make their decisions in private. Participants were explicitly assured that their decisions would remain confidential."

"After submitting their decisions, but before learning the outcomes of the game, participants responded to a post experimental questionnaire assessing their expectations regarding the behaviour of ingroup and outgroup members (i.e., how many chips, on average, ingroup and outgroup members had contributed to each of the pools)."

Differences from the Original Study

In the original study, participants were paid in NIS (new Israeli shekel). Because the replication's sample came from the University of Winchester in the UK, participants were paid in British Pound Sterling. On July 31, 2014 when the procedure was developed, one NIS was worth £0.17. For practical reasons, as there is a 20p (£0.2) coin, the £0.17 was rounded up to £0.20.

Additionally, the original study only included male participants. Because University of Winchester is heavily female dominated and has a small student body, collecting only male data would be much more difficult than including both sexes. A footnote in the original article indicated that in the IPD-MD game, there were "no differences between all-female and all-male groups" (Halevy et al., p. 407) suggesting that the inclusion of women in the study should not